

The oliGARCHic Transformation

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper defines and formalizes an *oliGARCHic transformation*: a structural transformation that begins with the classical nine-point construction of Euclidean triangle geometry and terminates in a nine-point configuration in $\mathbb{C}^4$ possessing cyclic, algebraic, metric, measure-theoretic, and dynamical structure. The construction is motivated by a general methodological principle: a finite geometric configuration may be transformed into an algebraic state space, lifted into a higher-dimensional representation, equipped with a metric and measure, and subsequently closed under a nonlinear dynamical system.

The classical nine-point circle supplies nine distinguished geometric states. Cyclicization associates those states with the cyclic group $\mathbb{Z}/9\mathbb{Z}$; roots of unity encode the resulting states; a Fourier-type lifting maps them into $\mathbb{C}^4$; the resulting nine points form a regular 8-simplex when $\mathbb{C}^4$ is regarded as an eight-dimensional real inner-product space; and a polynomial dynamical system is constructed whose fixed-point set consists of exactly those nine points. The framework is interpreted, cautiously and structurally, across geometry, algebra, analysis, statistics, economics, finance, physics, chemistry, biology, psychology, philosophy, political science, international relations, warfare economics, computer science, data science, and artificial intelligence.

The paper ends with “The End”

Contents

1 Introduction	3
2 The Classical Nine-Point Construction	3
3 Definition of the oliGARCHic Transformation	4
4 Stage I: Label Extraction	4
5 Stage II: Geometric Normalization	5
6 Stage III: Cyclicization	5
7 Stage IV: Fourier Lifting into $\mathbb{C}^4$	6
8 The Regular 8-Simplex Theorem	6
9 Normalization to the Unit Sphere	7
10 Stage V: Measure, Metric, Distance, and Dynamics	8
11 Stage VI: Dynamical Closure	8
12 The Complete Transformation Diagram	9

13 Algebraic and Group-Theoretic Interpretation	10
14 Geometric Interpretation	10
15 Statistical Interpretation	11
16 Econometric Interpretation	11
17 Finance and Risk Interpretation	12
18 Physics	12
19 Chemistry and Molecular-State Interpretation	13
20 Biology	13
21 Psychology and Psychiatry	13
22 Philosophical Interpretation	14
23 Religion and Symbolic Systems	14
24 Political Science, Geopolitics, and International Relations	14
25 Diplomacy and International Relations	15
26 Welfare Economics	15
27 Warfare Economics	15
28 Computer Science and Algorithms	16
29 Data Science and Representation Learning	16
30 Artificial Intelligence and Machine Learning	17
31 Statistics of the Nine-Point Configuration	18
32 A General n -Point Generalization	18
33 Comparison of the Two Nine-Point Geometries	19
34 Conceptual Interpretation Across Disciplines	19
35 Logical Structure of the Construction	20
36 Limitations	20
37 Conclusion	21

List of Figures

1 The oliGARCHic transformation from classical nine-point geometry to a nine-point dynamical system in $\mathbb{C}^4$	10
---	----

List of Tables

1	Classical and oliGARCHic nine-point constructions	19
2	Interpretive roles of the oliGARCHic construction	19

1 Introduction

The classical nine-point circle is one of the canonical constructions of Euclidean geometry. Given a non-degenerate triangle ABC , the midpoints of its three sides, the three feet of its altitudes, and the midpoints of the three segments joining the vertices to the orthocenter lie on a common circle [1, 2].

The present paper asks a different question. Suppose that the number nine is retained while the geometric environment is progressively transformed. Can the nine distinguished states of triangle geometry be converted into a canonical nine-state algebraic system, lifted into $\mathbb{C}^4$, endowed with a higher-dimensional metric, and finally realized as the fixed points of a polynomial dynamical system?

The answer is affirmative.

The principal chain is

$$ABC \longrightarrow \mathcal{N}_\Delta \longrightarrow \mathcal{N}_9^{\text{lab}} \longrightarrow C_9 \longrightarrow \{\zeta^k\}_{k=0}^8 \longrightarrow \mathcal{S}_9(\mathbb{C}^4) \longrightarrow \text{Fix}(T). \quad (1)$$

The transformation combines ideas from Euclidean geometry, finite cyclic groups, Fourier analysis, complex geometry, simplex theory, polynomial dynamics, measure theory, statistics, and computational mathematics [1, 3, 5, 6, 11].

2 The Classical Nine-Point Construction

Let ABC be a non-degenerate triangle with circumcenter O , orthocenter H , and circumradius R .

Define the three side midpoints by

$$M_{BC} = \frac{B+C}{2}, \quad M_{CA} = \frac{C+A}{2}, \quad M_{AB} = \frac{A+B}{2}. \quad (2)$$

Let D_A, D_B, D_C denote the feet of the altitudes from A, B, C , respectively. Finally define

$$M_{AH} = \frac{A+H}{2}, \quad M_{BH} = \frac{B+H}{2}, \quad M_{CH} = \frac{C+H}{2}. \quad (3)$$

The nine-point set is therefore

$$\mathcal{N}_\Delta = \{M_{BC}, M_{CA}, M_{AB}, D_A, D_B, D_C, M_{AH}, M_{BH}, M_{CH}\}. \quad (4)$$

Theorem 1 (Nine-point circle). *For every non-degenerate triangle ABC , the nine named points in $\mathcal{N}_\Delta$, counted with multiplicity when geometric points coincide, lie on a circle of radius $R/2$. Its center is the midpoint of O and H .*

The theorem is classical and follows from the standard affine and orthocentric geometry of the triangle [1, 2].

Remark 1. *For special triangles, particularly the equilateral triangle, some of the nine named points coincide. Thus “nine-point construction” is most precisely interpreted as a labeled nine-state construction, with multiplicity. The target $\mathbb{C}^4$ construction below has nine distinct points.*

3 Definition of the oliGARCHic Transformation

Definition 1 (oliGARCHic transformation). *An oliGARCHic transformation is the composition of structural transformations*

$$\mathfrak{D} = D \circ M \circ F \circ R \circ C \circ N \circ L, \quad (5)$$

where:

L, label extraction extracts the nine distinguished states of the classical construction;

N, normalization removes translation and scale by centering at the nine-point center and normalizing by the nine-point radius;

C, cyclicization replaces arbitrary angular coordinates by the canonical cyclic state space $\mathbb{Z}/9\mathbb{Z}$;

R, root-of-unity encoding represents the cyclic states by the ninth roots of unity;

F, Fourier lifting maps each root of unity into four complex coordinates;

M, metric reconstitution equips the lifted configuration with the Hermitian-Euclidean metric;

D, dynamical closure constructs a polynomial map having precisely the nine lifted points as fixed points.

The essential conceptual transformation is

$$\text{Euclidean incidence} \rightarrow \text{cyclic state space} \rightarrow \text{complex algebra} \rightarrow \text{higher-dimensional metric} \rightarrow \text{dynamics}. \quad (6)$$

4 Stage I: Label Extraction

The first stage deliberately discards geometric information that is not required to identify the nine-point construction.

Let

$$L(ABC) = (N_0, N_1, \dots, N_8) \quad (7)$$

be a fixed labeling of the nine classical states.

The labeling is not unique and therefore constitutes part of the model specification. An arbitrary nine-point triangle does not possess a canonical cyclic ordering equivalent to the ordering of the roots of unity.

The oliGARCHic transformation therefore preserves the nine-state cardinality while replacing the original geometric ordering with an explicitly declared algebraic ordering.

5 Stage II: Geometric Normalization

Let N denote the nine-point center. For each labeled point define

$$w_j = N_j - N. \quad (8)$$

Since the nine-point circle has radius $R/2$, define

$$u_j = \frac{2w_j}{R}. \quad (9)$$

Then every u_j lies on the unit circle:

$$|u_j| = 1. \quad (10)$$

Thus the original dimensional geometry is transformed into a dimensionless configuration on S^1 .

This step is analogous to standardization in statistics, nondimensionalization in physics, and normalization in numerical analysis. Removing irrelevant location and scale is often essential when the object of interest is structural rather than dimensional [8, 5, 15].

6 Stage III: Cyclicization

The nine-point circle supplies nine labeled states. We now replace their arbitrary angular positions by the canonical finite cyclic group

$$C_9 = \mathbb{Z}/9\mathbb{Z}. \quad (11)$$

The nine states are represented by

$$k = 0, 1, \dots, 8. \quad (12)$$

Let

$$\zeta = e^{2\pi i/9}. \quad (13)$$

Then

$$\zeta^9 = 1 \quad (14)$$

and

$$\zeta^k \neq 1, \quad k = 1, \dots, 8. \quad (15)$$

The cyclic encoding is therefore

$$k \longmapsto \zeta^k. \quad (16)$$

The distinction between the original and transformed systems is crucial:

$$\text{classical system : nine geometrically determined positions,} \quad (17)$$

$$\text{cyclic system : nine algebraically indexed states.} \quad (18)$$

No assertion is made that the nine classical points are equally spaced on the nine-point circle.

7 Stage IV: Fourier Lifting into $\mathbb{C}^4$

Define the lifting map

$$\Phi : C_9 \longrightarrow \mathbb{C}^4, \quad (19)$$

$$\Phi(k) = \begin{pmatrix} \zeta^k \\ \zeta^{2k} \\ \zeta^{3k} \\ \zeta^{4k} \end{pmatrix}. \quad (20)$$

Write

$$p_k = \Phi(k) = (\zeta^k, \zeta^{2k}, \zeta^{3k}, \zeta^{4k}), \quad k = 0, \dots, 8. \quad (21)$$

The construction is Fourier-like because its coordinates are characters of the finite cyclic group C_9 [3, 4].

Each coordinate satisfies

$$|\zeta^{jk}| = 1, \quad (22)$$

so

$$\|p_k\|^2 = \sum_{j=1}^4 |\zeta^{jk}|^2 = 4. \quad (23)$$

Consequently,

$$\|p_k\| = 2. \quad (24)$$

The nine points lie on the sphere of radius 2 in $\mathbb{C}^4$.

8 The Regular 8-Simplex Theorem

The central geometric result is the following.

Theorem 2 (Regular nine-point configuration in $\mathbb{C}^4$). *The points*

$$p_k = (\zeta^k, \zeta^{2k}, \zeta^{3k}, \zeta^{4k}), \quad k = 0, \dots, 8, \quad (25)$$

form the vertices of a regular 8-simplex when $\mathbb{C}^4$ is regarded as an eight-dimensional real inner-product space.

Proof. Let $k \neq \ell$ and put $d = k - \ell$. The standard Hermitian inner product gives

$$\langle p_k, p_\ell \rangle = \sum_{j=1}^4 \zeta^{jd}. \quad (26)$$

Because $d \not\equiv 0 \pmod{9}$,

$$\sum_{j=0}^8 \zeta^{jd} = 0. \quad (27)$$

The terms indexed by j and $9 - j$ are complex conjugates. Hence

$$2 \operatorname{Re} \left(\sum_{j=1}^4 \zeta^{jd} \right) = -1, \quad (28)$$

and therefore

$$\operatorname{Re} \langle p_k, p_\ell \rangle = -\frac{1}{2}. \quad (29)$$

Consequently,

$$\|p_k - p_\ell\|^2 = \|p_k\|^2 + \|p_\ell\|^2 - 2 \operatorname{Re} \langle p_k, p_\ell \rangle \quad (30)$$

$$= 4 + 4 - 2 \left(-\frac{1}{2} \right) \quad (31)$$

$$= 9. \quad (32)$$

Thus

$$\|p_k - p_\ell\| = 3 \quad (33)$$

for every $k \neq \ell$. Therefore all nine points are mutually equidistant and form a regular 8-simplex. $\square$

Remark 2. *The full complex inner product is not generally equal to -1 . The invariant required for Euclidean distance is its real part, which is $-1/2$. This distinction is essential because $\mathbb{C}^4$ is being interpreted as $\mathbb{R}^8$ for the simplex geometry.*

9 Normalization to the Unit Sphere

Define

$$q_k = \frac{1}{2} p_k. \quad (34)$$

Then

$$\|q_k\| = 1 \quad (35)$$

and

$$\|q_k - q_\ell\| = \frac{3}{2}, \quad k \neq \ell. \quad (36)$$

Hence the normalized configuration is a regular 8-simplex inscribed in

$$S^7 \subset \mathbb{R}^8 \cong \mathbb{C}^4. \quad (37)$$

The construction therefore converts a circle-based nine-state geometry into a maximally symmetric finite configuration on the seven-sphere.

10 Stage V: Measure, Metric, Distance, and Dynamics

The appropriate unified object is not a claim that a measure, metric, and distance are literally identical mathematical types. Rather, it is a structured object carrying all three structures simultaneously.

Definition 2 (Nine-point UMD). *Define the nine-point Unified Measure–Metric–Distance object by*

$$\mathfrak{U}_9 = (\mathbb{C}^4, \mathcal{B}, \mu_8, d, T), \quad (38)$$

where $\mathcal{B}$ is the Borel σ -algebra, μ_8 is eight-dimensional Lebesgue measure, d is the Hermitian-Euclidean metric, and T is the polynomial dynamical map defined below.

The metric is

$$d(z, w) = \left(\sum_{j=1}^4 |z_j - w_j|^2 \right)^{1/2}. \quad (39)$$

The measure and metric are linked through the standard identification

$$\mathbb{C}^4 \cong \mathbb{R}^8 \quad (40)$$

and through the usual relationship between metric length and one-dimensional Hausdorff measure on rectifiable curves [7].

11 Stage VI: Dynamical Closure

Define

$$F : \mathbb{C}^4 \longrightarrow \mathbb{C}^4, \quad (41)$$

$$F(z_1, z_2, z_3, z_4) = \begin{pmatrix} z_1^9 - 1 \\ z_2 - z_1^2 \\ z_3 - z_1^3 \\ z_4 - z_1^4 \end{pmatrix}. \quad (42)$$

For any $\lambda \in \mathbb{C} \setminus \{0\}$, define

$$T(z) = z + \lambda F(z). \quad (43)$$

Theorem 3 (Exact nine-point fixed-point theorem). *The fixed-point set of T is precisely*

$$\text{Fix}(T) = \left\{ (\zeta^k, \zeta^{2k}, \zeta^{3k}, \zeta^{4k}) : k = 0, \dots, 8 \right\}. \quad (44)$$

Proof. A fixed point satisfies

$$T(z) = z. \quad (45)$$

Since $\lambda \neq 0$, this is equivalent to

$$F(z) = 0. \quad (46)$$

Hence

$$z_1^9 = 1, \quad (47)$$

$$z_2 = z_1^2, \quad (48)$$

$$z_3 = z_1^3, \quad (49)$$

$$z_4 = z_1^4. \quad (50)$$

The first equation has exactly nine complex solutions,

$$z_1 = \zeta^k, \quad k = 0, \dots, 8. \quad (51)$$

The remaining equations uniquely determine the other coordinates. Thus exactly nine fixed points exist. $\square$

This establishes the desired correspondence:

$$\text{nine geometric states} \longrightarrow \text{nine cyclic states} \quad (52)$$

$$\longrightarrow \text{nine algebraic states} \quad (53)$$

$$\longrightarrow \text{nine metric states} \quad (54)$$

$$\longrightarrow \text{nine dynamical fixed points.} \quad (55)$$

12 The Complete Transformation Diagram

Figure 1 summarizes the transformation.

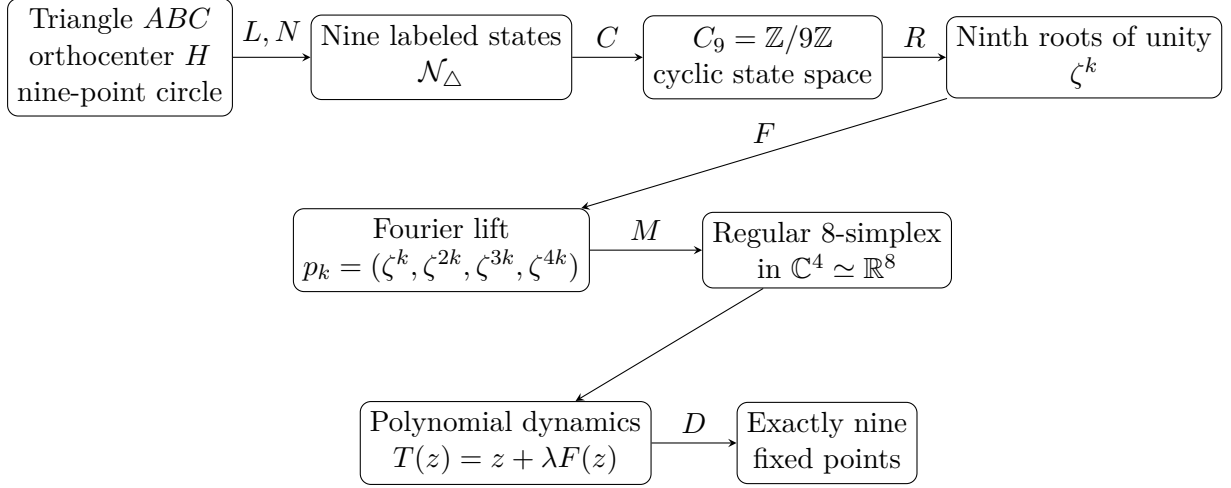


Figure 1: The oliGARCHic transformation from classical nine-point geometry to a nine-point dynamical system in $\mathbb{C}^4$.

13 Algebraic and Group-Theoretic Interpretation

The roots of unity form the cyclic group

$$C_9 = \langle \zeta : \zeta^9 = 1 \rangle. \quad (56)$$

The coordinate functions

$$\chi_j(k) = \zeta^{jk}, \quad j = 1, \dots, 4, \quad (57)$$

are characters of the finite cyclic group. The map

$$k \mapsto (\chi_1(k), \chi_2(k), \chi_3(k), \chi_4(k)) \quad (58)$$

is therefore a representation-theoretic embedding of the nine-state cyclic system into $\mathbb{C}^4$.

The construction illustrates a general principle of harmonic analysis:

$$\text{finite symmetry} \longrightarrow \text{characters} \longrightarrow \text{coordinate representation}. \quad (59)$$

The same general architecture underlies discrete Fourier transforms, spectral methods, convolutional algorithms, and many signal-processing systems [3, 4, 12].

14 Geometric Interpretation

The classical nine-point circle is essentially two-dimensional. The target configuration has real dimension eight.

The dimension identity is

$$\dim_{\mathbb{R}} \mathbb{C}^4 = 8. \quad (60)$$

A regular simplex with nine vertices has affine dimension eight. Thus $\mathbb{C}^4$ is exactly large enough to contain a regular 8-simplex.

This gives the dimensional correspondence

$$9 \text{ states} \longleftrightarrow 8 \text{ independent real directions} \quad (61)$$

$$\longleftrightarrow \mathbb{C}^4. \quad (62)$$

The transformation is therefore not an arbitrary dimensional enlargement. It reaches the minimal real dimension required for nine mutually equidistant points.

15 Statistical Interpretation

The nine states can be regarded as a finite state space

$$\mathcal{X}_9 = \{0, 1, \dots, 8\}. \quad (63)$$

A probability distribution is a vector

$$\pi = (\pi_0, \dots, \pi_8), \quad \pi_k \geq 0, \quad \sum_{k=0}^8 \pi_k = 1. \quad (64)$$

The uniform distribution

$$\pi_k = \frac{1}{9} \quad (65)$$

is invariant under the cyclic permutation

$$k \longmapsto k + 1 \pmod{9}. \quad (66)$$

The Fourier coordinates diagonalize cyclic shifts, which is one reason spectral representations are useful in statistical signal processing and stochastic systems [3, 13].

The nine-point construction can therefore be interpreted as a finite-state statistical model whose state geometry has an explicitly symmetric embedding.

16 Econometric Interpretation

In econometrics, the nine states can be interpreted as regimes,

$$S_t \in \{0, \dots, 8\}. \quad (67)$$

The cyclic structure permits a transition operator P satisfying

$$P_{ij} = p_{j-i \pmod{9}}. \quad (68)$$

Such a transition matrix is circulant and is diagonalized by the discrete Fourier basis.

This connects the construction to regime-switching models, Markov chains, state-space methods, and spectral econometrics [14, 15].

A generic observation equation could be written as

$$Y_t = \alpha_{S_t} + \beta'_{S_t} X_t + \varepsilon_t. \quad (69)$$

The nine-point geometry does not itself imply this econometric model. Rather, it supplies a structured state space upon which such a model can be defined.

17 Finance and Risk Interpretation

A financial application can interpret the nine states as regimes of risk, liquidity, volatility, or market structure.

For example,

$$R_t = \mu_{S_t} + \sigma_{S_t} \varepsilon_t \quad (70)$$

allows regime-dependent conditional volatility.

The cyclic representation provides a spectral basis for analyzing transitions between regimes, while the $\mathbb{C}^4$ embedding provides a geometric representation of state similarity.

A stylized risk functional may be written as

$$\mathcal{R}(\pi) = \sum_{k=0}^8 \pi_k r_k. \quad (71)$$

The construction itself does not establish an empirical financial law. Its contribution is representational: it supplies a finite symmetric state architecture within which financial regime models may be specified and tested.

18 Physics

The transformation has a natural interpretation in statistical mechanics and dynamical systems.

The nine states may be regarded as discrete configurations, while T defines a deterministic nonlinear evolution.

The fixed points satisfy

$$T(z) = z. \quad (72)$$

In a dynamical-systems interpretation, these are equilibrium states. Their local stability would be determined by the eigenvalues of the Jacobian

$$J_T(z^*) = I + \lambda J_F(z^*). \quad (73)$$

No stability claim follows merely from the existence of the nine fixed points. Stability requires a separate spectral analysis [9, 10].

19 Chemistry and Molecular-State Interpretation

A finite-state molecular model can regard the nine states as conformational, electronic, or reaction-network states. The transition structure can then be represented by a graph or Markov generator.

For a continuous-time model,

$$\frac{d\pi(t)}{dt} = \pi(t)Q, \quad (74)$$

where Q is a generator satisfying

$$Q_{ij} \geq 0, \quad i \neq j, \quad (75)$$

and

$$\sum_j Q_{ij} = 0. \quad (76)$$

The analogy is structural rather than chemical: the mathematical object provides a finite configuration space, while physical chemistry supplies the empirical interpretation [16].

20 Biology

In mathematical biology, nine states may represent phenotypic, cellular, ecological, or epidemiological regimes.

A discrete nonlinear model may be written

$$x_{t+1} = G(x_t, S_t). \quad (77)$$

Alternatively, the nine-point states may represent coarse-grained attractor candidates in a high-dimensional biological state space.

This is consistent with the broader use of dynamical systems, state-space reduction, and nonlinear models in mathematical biology [17].

21 Psychology and Psychiatry

The same architecture can be used only as a mathematical state-space metaphor, not as a diagnostic theory.

For example, a latent-state model could write

$$S_t \in \{0, \dots, 8\}, \quad (78)$$

with observed behavioral variables

$$Y_t = g(S_t, X_t, \varepsilon_t). \quad (79)$$

The distinction between latent state, observed variable, transition dynamics, and measurement process is fundamental to state-space modeling.

No psychiatric diagnosis or clinical conclusion follows from the nine-point construction.

22 Philosophical Interpretation

The transformation separates three notions that are often conflated:

$$\text{identity} \neq \text{representation} \neq \text{dynamics}. \quad (80)$$

A classical nine-point state has an identity supplied by Euclidean geometry. The root-of-unity state supplies an algebraic representation. The $\mathbb{C}^4$ vector supplies a metric representation. The polynomial map supplies a dynamical realization.

Thus one object may possess several mathematically legitimate representations without those representations being identical.

This distinction echoes the broader philosophical difference between an object, its representation, and the laws governing transformations of the representation.

23 Religion and Symbolic Systems

The number nine has appeared in numerous cultural and religious symbolic systems. Such symbolism, however, should not be confused with the mathematical properties established here.

The mathematical significance of nine in this paper is structural:

$$9 = 8 + 1, \quad (81)$$

and nine mutually equidistant points require eight real dimensions.

Thus the mathematical nine is a consequence of simplex geometry and finite cyclic algebra rather than a theological assertion.

24 Political Science, Geopolitics, and International Relations

The nine-state representation can be interpreted as a finite strategic state space.

Let

$$S_t \in C_9 \quad (82)$$

denote a geopolitical regime. A transition kernel may then be written

$$P(S_{t+1} = j \mid S_t = i) = P_{ij}. \quad (83)$$

The cyclic representation is useful when the states have an ordered or rotational interpretation.

For strategic interaction, define payoff functions

$$u_i(s_i, s_{-i}). \quad (84)$$

The nine-point geometry does not impose a particular equilibrium concept. Nash equilibrium, correlated equilibrium, bargaining solutions, and repeated-game dynamics remain separate theoretical constructions [18, 19, 20].

The advantage of the oliGARCHic representation is that strategic states can be embedded into a common metric space.

25 Diplomacy and International Relations

Diplomatic systems frequently involve several simultaneous dimensions: cooperation, competition, security, trade, information, and institutional coordination.

A vector representation

$$z_k \in \mathbb{C}^4 \quad (85)$$

can therefore be interpreted as a multidimensional state representation.

The regular-simplex property implies that the nine canonical states are equidistant in the chosen metric. This should be interpreted as a mathematical symmetry assumption, not as an empirical statement that real states or diplomatic positions are equally distant.

26 Welfare Economics

Suppose the nine states have welfare values

$$W_k. \quad (86)$$

A probability-weighted welfare index is

$$\overline{W} = \sum_{k=0}^8 \pi_k W_k. \quad (87)$$

Different welfare criteria can then be imposed on the same state space.

For example, a concave social welfare function may be written

$$\mathcal{W} = \sum_{k=0}^8 \pi_k u(W_k). \quad (88)$$

The geometry and the welfare criterion remain conceptually distinct. The former describes the state space; the latter evaluates states [21].

27 Warfare Economics

A finite nine-state system can also represent strategic escalation regimes.

Let

$$S_t \in \{\text{peace, deterrence, mobilization, } \dots\}. \quad (89)$$

A transition model can represent movement among regimes, while an economic loss function may be written

$$L(S_t, S_{t+1}, X_t). \quad (90)$$

A policy problem may then take the form

$$\min_{\pi} \mathbb{E} \left[\sum_{t=0}^T \delta^t L(S_t, S_{t+1}, X_t) \right]. \quad (91)$$

This connects naturally to dynamic programming, optimal control, deterrence theory, and strategic interaction [18, 22].

The mathematical construction does not advocate warfare. It provides a possible state-space architecture for analyzing transitions among strategic regimes.

28 Computer Science and Algorithms

The nine-point system has an exceptionally simple computational representation.

INPUT:

```
primitive ninth root zeta
nonzero complex parameter lambda
```

FOR $k = 0, \dots, 8$:

```
p[k] = (zeta^k, zeta^(2k), zeta^(3k), zeta^(4k))
```

DEFINE $F(z)$:

```
return (
    z1^9 - 1,
    z2 - z1^2,
    z3 - z1^3,
    z4 - z1^4
)
```

DEFINE $T(z)$:

```
return z + lambda * F(z)
```

VERIFY:

```
T(p[k]) = p[k] for every k
```

VERIFY DISTINCTNESS:

```
p[k] != p[l] whenever k != l
```

VERIFY METRIC SYMMETRY:

```
distance(p[k], p[l]) = 3 for k != l
```

The construction requires only a finite loop over nine states. Symbolically, the generation of the complete fixed-point set is therefore computationally inexpensive.

29 Data Science and Representation Learning

The mapping

$$k \longmapsto p_k \in \mathbb{C}^4 \quad (92)$$

can be interpreted as a deterministic embedding of categorical states into a continuous vector space.

Unlike an arbitrary learned embedding, however, this embedding has exact algebraic structure:

$$\|p_k\| = 2, \tag{93}$$

$$\|p_k - p_\ell\| = 3, \quad k \neq \ell. \tag{94}$$

This makes the construction a useful benchmark for algorithms that claim to preserve categorical symmetry, pairwise distances, or cyclic structure.

A representation-learning objective could therefore include

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \alpha \mathcal{L}_{\text{geometry}} + \beta \mathcal{L}_{\text{symmetry}}. \tag{95}$$

The geometric terms would penalize departures from the exact simplex relations.

30 Artificial Intelligence and Machine Learning

The nine-state construction provides a controlled test environment for several machine-learning problems:

1. categorical embedding;
2. manifold representation;
3. cyclic state prediction;
4. graph learning;
5. spectral learning;
6. nonlinear dynamical-system identification;
7. fixed-point detection;
8. symmetry-preserving representation learning.

For a learned transition function G_θ , one could impose

$$\mathcal{L}_{\text{fixed}} = \sum_{k=0}^8 \|G_\theta(p_k) - p_k\|^2. \tag{96}$$

A perfect realization satisfies

$$\mathcal{L}_{\text{fixed}} = 0. \tag{97}$$

The construction is therefore a useful synthetic benchmark because the ground truth is known exactly.

31 Statistics of the Nine-Point Configuration

Let

$$\bar{p} = \frac{1}{9} \sum_{k=0}^8 p_k. \quad (98)$$

For every coordinate,

$$\sum_{k=0}^8 \zeta^{jk} = 0, \quad j = 1, \dots, 4. \quad (99)$$

Therefore

$$\bar{p} = 0. \quad (100)$$

The configuration is centered.

Its empirical covariance, after identifying $\mathbb{C}^4$ with $\mathbb{R}^8$, has a highly symmetric form. Consequently, the configuration provides a natural finite benchmark for isotropy, symmetry detection, principal-component analysis, clustering, and multidimensional scaling.

The exact regular-simplex structure is particularly useful for testing whether an algorithm introduces spurious anisotropy.

32 A General n -Point Generalization

The nine-point construction is one member of a broader family.

Let $n \geq 2$ and let

$$\zeta_n = e^{2\pi i/n}. \quad (101)$$

When $n = 2m + 1$ is odd, define

$$p_k^{(n)} = (\zeta_n^k, \zeta_n^{2k}, \dots, \zeta_n^{mk}) \in \mathbb{C}^m. \quad (102)$$

Since

$$\dim_{\mathbb{R}} \mathbb{C}^m = 2m = n - 1, \quad (103)$$

this is precisely the minimal real dimension required for a regular simplex with n vertices. For $n = 9$,

$$m = 4, \quad (104)$$

and the construction becomes

$$p_k = (\zeta^k, \zeta^{2k}, \zeta^{3k}, \zeta^{4k}). \quad (105)$$

Thus the nine-point construction is not an isolated curiosity. It belongs to a systematic family of cyclic Fourier simplex embeddings.

33 Comparison of the Two Nine-Point Geometries

Table 1: Classical and oliGARCHic nine-point constructions

Property	Classical construction	oliGARCHic construction
Primary space	$\mathbb{R}^2$	$\mathbb{C}^4 \simeq \mathbb{R}^8$
Generating object	Triangle	Cyclic group C_9
Number of states	Nine, with multiplicity	Nine distinct states
Primary geometry	Circle	Regular 8-simplex
Symmetry	Triangle-dependent	Exact cyclic symmetry
Normalization	Radius $R/2$	Unit sphere S^7
Coordinates	Euclidean	Complex Fourier coordinates
Pairwise distances	Generally unequal	Exactly 3
Dynamics	Not intrinsic	Polynomial fixed-point system
Measure	Planar	Eight-dimensional
State interpretation	Geometric	Algebraic and dynamical

The distinction can be summarized as

$$\text{nine points on a circle} \longrightarrow \text{nine vertices of a regular simplex.} \quad (106)$$

The transformation therefore preserves cardinality while radically changing geometry.

34 Conceptual Interpretation Across Disciplines

Table 2: Interpretive roles of the oliGARCHic construction

Field	Structural interpretation
Euclidean geometry	Nine-point circle
Algebra	Ninth roots of unity
Group theory	Cyclic group C_9
Harmonic analysis	Character/Fourier embedding
Complex analysis	Polynomial map on $\mathbb{C}^4$
Metric geometry	Regular 8-simplex
Measure theory	Lebesgue and Hausdorff measures
Dynamical systems	Nine exact fixed points
Statistics	Symmetric finite-state sample space
Econometrics	Nine-regime state-space model
Finance	Risk/regime representation
Physics	Equilibrium-state architecture
Chemistry	Finite molecular-state representation
Biology	Discrete dynamical state space
Psychology	Latent-state representation
Political science	Strategic regime space
International relations	Multidimensional state representation
Welfare economics	State-dependent welfare evaluation

Continued on next page

Table 2 – continued from previous page

Field	Structural interpretation
Warfare economics	Strategic escalation regimes
Computer science	Finite exact algorithm
Data science	Structured embedding benchmark
AI/ML	Symmetry-preserving representation learning

These interpretations should be understood as mathematical analogies or modeling architectures, not as empirical claims.

35 Logical Structure of the Construction

The logical dependencies can be summarized by the following implications.

$$ABC \text{ non-degenerate} \implies \mathcal{N}_\Delta \text{ exists,} \quad (107)$$

$$\mathcal{N}_\Delta \implies 9 \text{ labeled states,} \quad (108)$$

$$9 \text{ states} \implies C_9 \text{ state indexing,} \quad (109)$$

$$C_9 \implies 9 \text{ roots of unity,} \quad (110)$$

$$\{\zeta^k\} \implies \{p_k\} \subset \mathbb{C}^4, \quad (111)$$

$$\{p_k\} \implies \text{regular 8-simplex,} \quad (112)$$

$$\{p_k\} \implies \text{Fix}(T) = \{p_k\}. \quad (113)$$

The reverse implications generally do not hold. In particular, the existence of nine equidistant points in $\mathbb{C}^4$ does not imply the existence of a particular triangle whose nine-point circle generated them.

Thus the oliGARCHic transformation is directional.

36 Limitations

Several distinctions are necessary.

First, the transformation is not an isometry between the classical nine-point circle and the $\mathbb{C}^4$ simplex.

Second, cyclicization is an encoding choice. The classical nine-point circle does not canonically determine the cyclic ordering used by the root-of-unity representation.

Third, the nine-point theorem is geometric, whereas the $\mathbb{C}^4$ construction is algebraic. The transformation therefore preserves the number and labeling of states rather than every geometric relation.

Fourth, the polynomial map T has exactly nine fixed points, but those fixed points are not automatically attracting. Stability depends on the Jacobian and the parameter λ .

Fifth, interdisciplinary interpretations are modeling possibilities. The mathematical construction itself contains no empirical economic, psychological, geopolitical, biological, or physical data.

These distinctions prevent category errors while preserving the usefulness of the common architecture.

37 Conclusion

The oliGARCHic transformation provides a formal bridge between two apparently unrelated nine-point constructions.

The classical triangle gives

$$\text{triangle} \longrightarrow \text{nine-point circle.} \quad (114)$$

The oliGARCHic transformation continues

$$\text{nine-point circle} \longrightarrow \text{nine labeled states} \quad (115)$$

$$\longrightarrow C_9 \quad (116)$$

$$\longrightarrow \text{ninth roots of unity} \quad (117)$$

$$\longrightarrow \mathbb{C}^4 \quad (118)$$

$$\longrightarrow \text{regular 8-simplex} \quad (119)$$

$$\longrightarrow \text{nine exact fixed points.} \quad (120)$$

The resulting object

$$\mathfrak{U}_9 = (\mathbb{C}^4, \mathcal{B}, \mu_8, d, T) \quad (121)$$

simultaneously supports measure-theoretic, metric, geometric, algebraic, statistical, and dynamical descriptions.

Its most striking feature is the coincidence

$$9 - 1 = 8 = \dim_{\mathbb{R}} \mathbb{C}^4, \quad (122)$$

which permits nine mutually equidistant states to occupy the minimal real dimension required for a regular 8-simplex.

The construction consequently demonstrates a general methodological principle:

$\text{finite geometric structure} \rightarrow \text{finite algebraic structure} \rightarrow \text{metric representation} \rightarrow \text{dynamical system.}$

(123)

The nine-point circle and the $\mathbb{C}^4$ nine-point simplex are therefore not the same object. The oliGARCHic transformation is precisely the mathematical operation that relates them while making explicit which structures are preserved, which are replaced, and which are newly introduced.

References

- [1] H. S. M. Coxeter, *Introduction to Geometry*, 2nd ed., Wiley, 1969.
- [2] R. A. Johnson, *Advanced Euclidean Geometry*, Dover, 2007.
- [3] A. Terras, *Fourier Analysis on Finite Groups and Applications*, Cambridge University Press, 1999.

- [4] E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*, Princeton University Press, 2003.
- [5] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.
- [6] F. Hirsch, *Social Limits to Growth*, Harvard University Press, Cambridge, MA, 1976.
- [7] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, American Mathematical Society, 2001.
- [8] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989.
- [9] S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed., Westview Press, 2015.
- [10] M. W. Hirsch, S. Smale, and R. L. Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*, 3rd ed., Academic Press, 2013.
- [11] C. Villani, *Optimal Transport: Old and New*, Springer, 2009.
- [12] J. W. Cooley and J. W. Tukey, “An algorithm for the machine calculation of complex Fourier series,” *Mathematics of Computation*, 19, 297–301, 1965.
- [13] C. E. Shannon, “A mathematical theory of communication,” *Bell System Technical Journal*, 27, 379–423 and 623–656, 1948.
- [14] J. D. Hamilton, *Time Series Analysis*, Princeton University Press, 1994.
- [15] F. Hayashi, *Econometrics*, Princeton University Press, 2000.
- [16] P. Atkins and J. de Paula, *Physical Chemistry*, Oxford University Press.
- [17] J. D. Murray, *Mathematical Biology*, 3rd ed., Springer, 2002.
- [18] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, 1960.
- [19] M. J. Osborne and A. Rubinstein, *A Course in Game Theory*, MIT Press, 1994.
- [20] K. N. Waltz, *Theory of International Politics*, McGraw–Hill, 1979.
- [21] A. Sen, *Collective Choice and Social Welfare*, Harvard University Press, 1970.
- [22] R. H. Strotz, “Myopia and inconsistency in dynamic utility maximization,” *Review of Economic Studies*, 23, 165–180, 1955.

Glossary

Affine dimension

The dimension of the smallest affine subspace containing a geometric configuration. Nine affinely independent points have affine dimension eight.

Cyclicization

The replacement of an arbitrary finite ordering of states by a canonical cyclic state space such as $\mathbb{Z}/9\mathbb{Z}$.

Fourier lifting

The mapping of a cyclic state k into a vector of character values. In the present construction this vector is

$$\Phi(k) = (\zeta^k, \zeta^{2k}, \zeta^{3k}, \zeta^{4k}). \quad (124)$$

Hermitian metric

The Euclidean metric obtained from the standard Hermitian inner product on $\mathbb{C}^4$.

Nine-point circle

The circle through the three side midpoints, three altitude feet, and three vertex–orthocenter midpoints of a non-degenerate triangle.

OliGARCHic transformation

The structural transformation defined in this paper from the classical nine-point construction to the cyclic, Fourier, metric, and dynamical nine-point construction in $\mathbb{C}^4$.

Regular simplex

A simplex whose vertices are mutually equidistant. Nine vertices therefore form a regular 8-simplex.

Root of unity

A complex number ζ satisfying $\zeta^n = 1$. A primitive ninth root of unity generates the cyclic group C_9 .

State space

The set of possible states of a mathematical or dynamical system.

UMD

Unified Measure–Metric–Distance object: a structured mathematical object carrying a measure, metric, distance function, and, in the present construction, dynamics.

Fixed point

A point z^* satisfying

$$T(z^*) = z^*. \tag{125}$$

The End